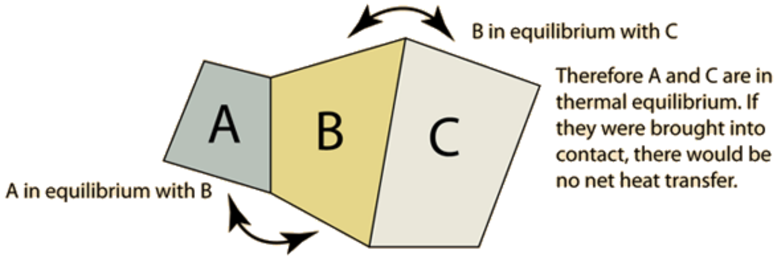
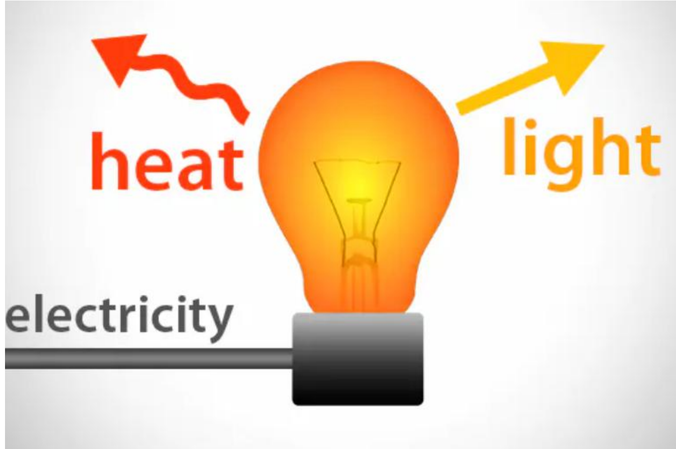
How is life sustained in an ecosystem

- It sustains life systems and ecological processes by energy flow
- Synthesis of organic matter by producers, i.e, productivity
- Decomposition of dead organic matter
- Cycling of minerals and nutrients

Energy is needed for us to do all kinds of work.

Thermodynamic laws provide the framework for bioenergetics



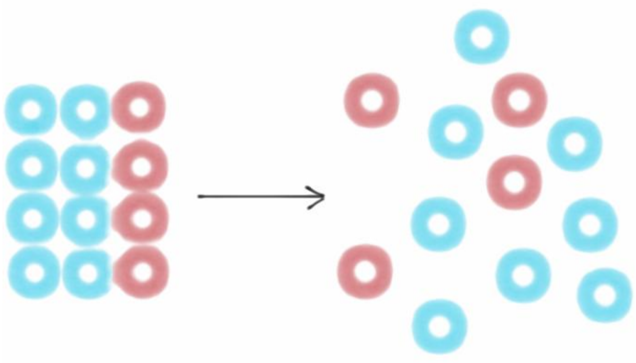


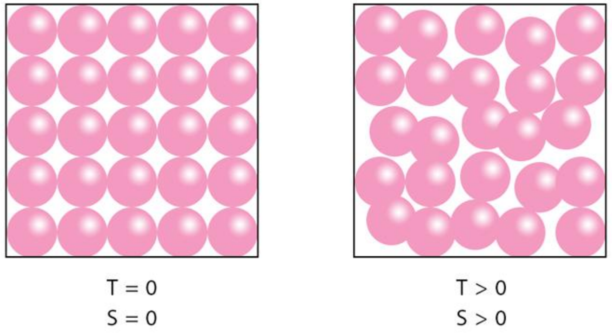
0. Thermal Equilibrium

1. Energy conservation

2. ENTROPY expansion

3. Absolute temperature



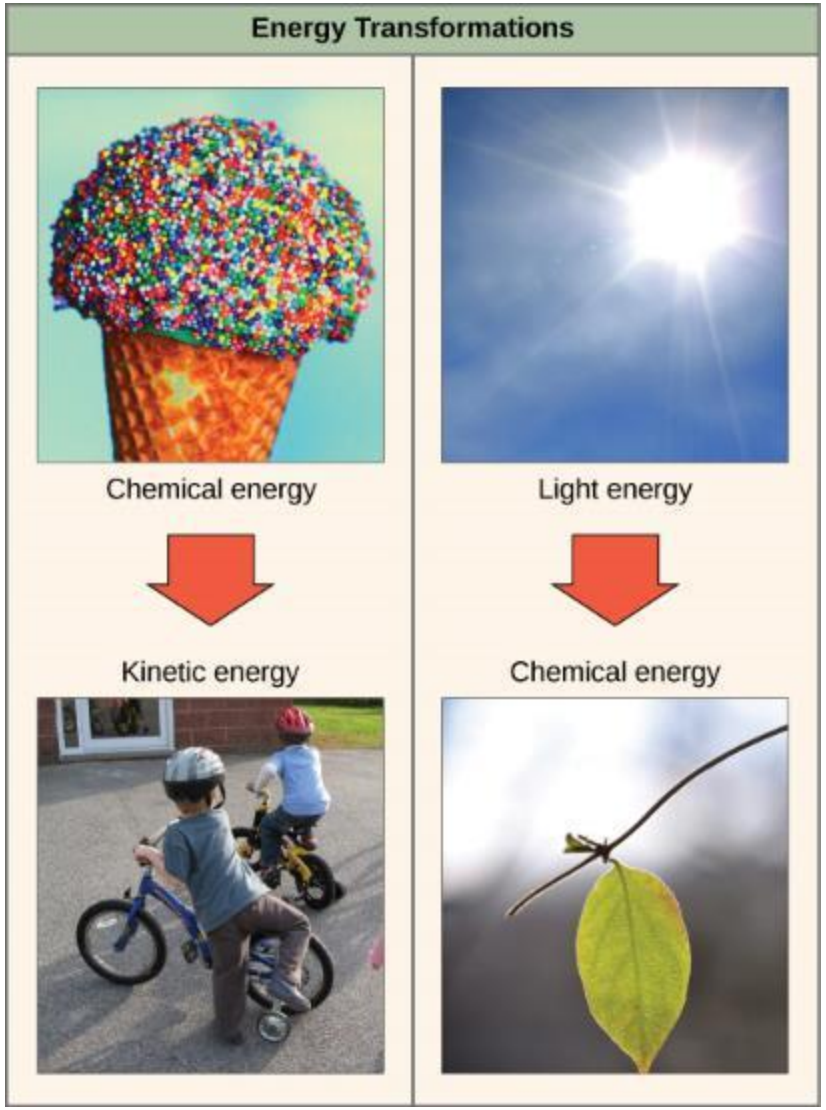


Energy transformations

Cellular energy can be *potential* or *kinetic.*

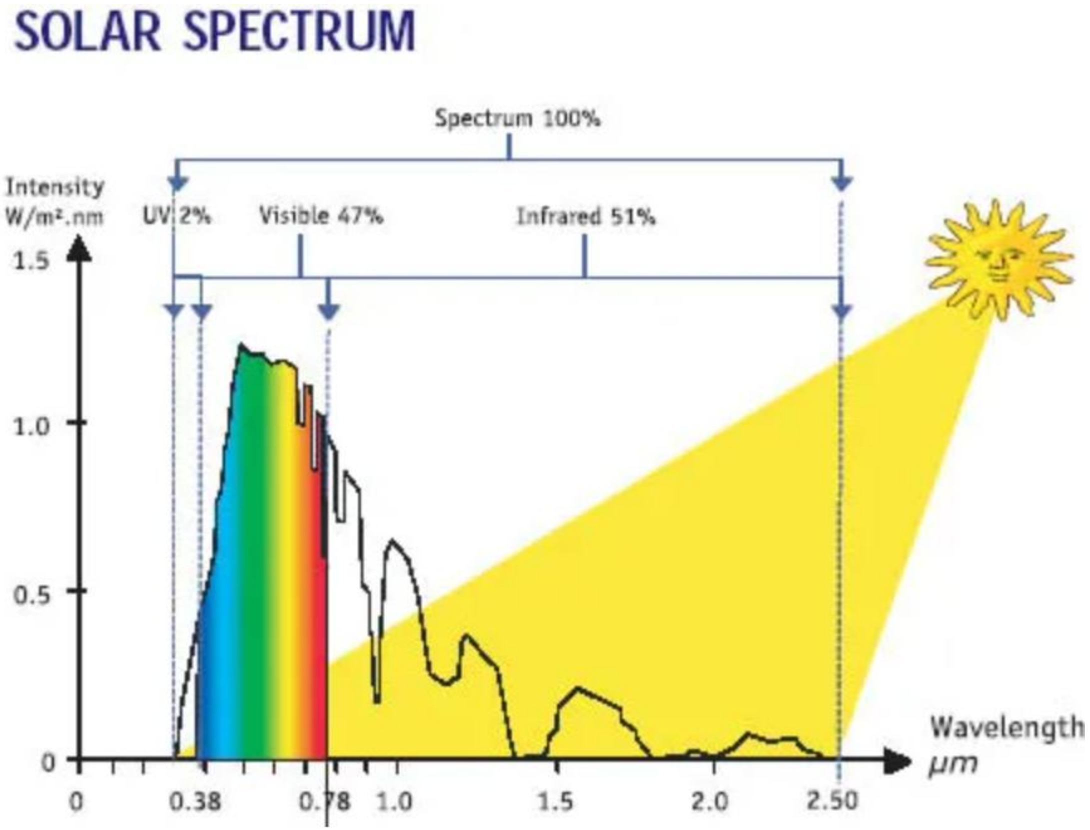
Potential energy is stationary, and it exists as chemical bonds, concentration gradients, and electrical charges in the cells.

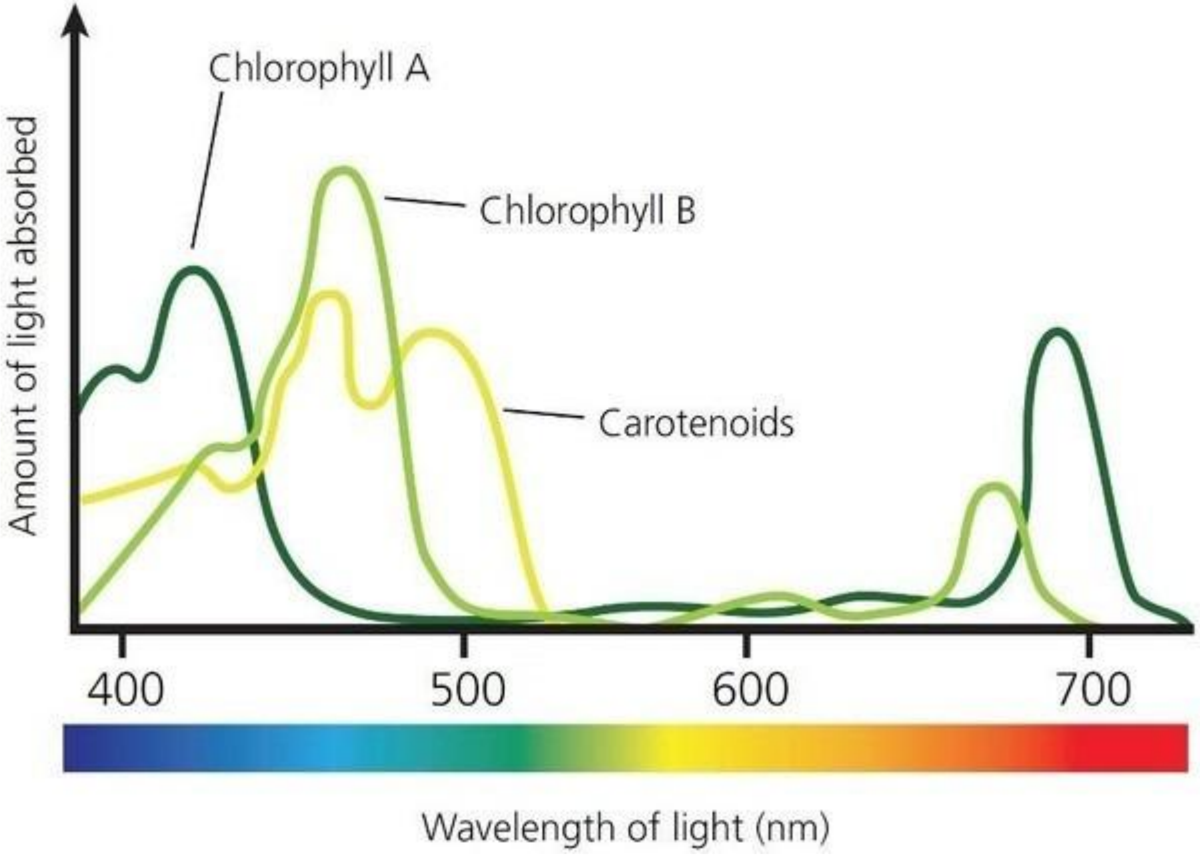
Kinetic energy is energy in motion, which results in the movement of molecules. Heat or thermal energy, for example, causes molecules to oscillate. In the cells, kinetic energy is converted from potential energy, moving electrons, hydrogen ions, and other charged particles



Source of energy

- Less than 50 percent of the incident solar radiation is photosynthetically active radiation
- Plants and photosynthetic bacteria capture 2-10 % of the photosynthetic active radiation and this small amount of energy sustains the entire living world.





Fundamental laws governing energy flow

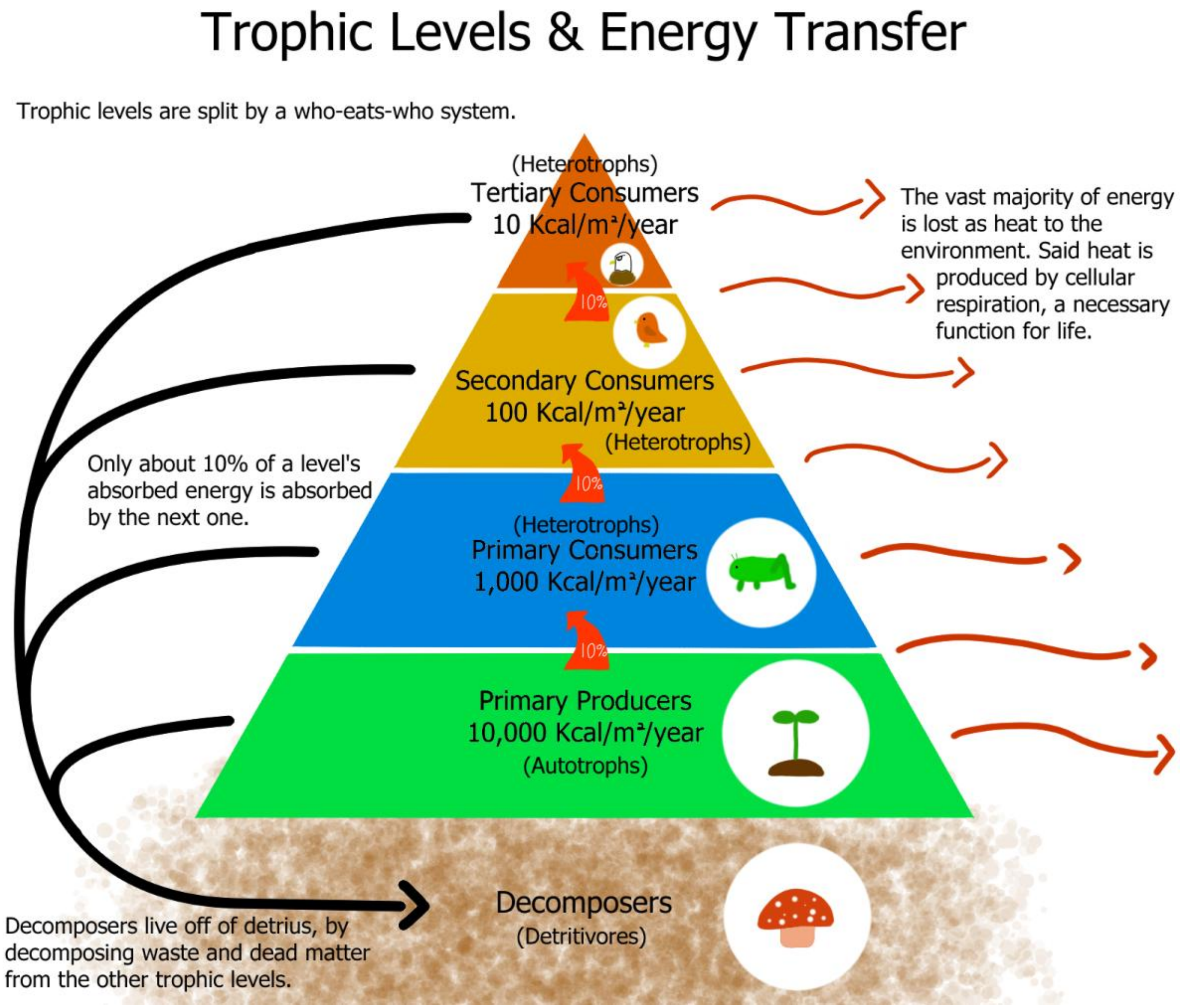
First law of thermodynamics

It states that energy can neither be created nor be destroyed. It can only be transferred from one form to another.

Hence the total energy of the universe is constant. During photosynthesis, there is a transformation of light energy (sunlight) into chemical energy (food).

Second law of thermodynamics

The energy transformation is not hundred percent efficient. So, some energy is dissipated as heat energy into the surroundings, while transforming from one form to another. Hence while energy gets transferred from one trophic level to another, some part of the energy is lost as heat, which cannot be reused. Therefore, energy flow in an ecosystem is unidirectional or one way flow from producers through herbivores to carnivores. So the energy can be used only once and cannot be recycled.



The energy available for the individuals at higher trophic levels will be gradually declining. Hence after 5th trophic level, the available energy will be insufficient for survival. This is the reason why the trophic levels are limited.

Bioenergetics

- Describes the transfer and utilization of energy in biologic systems. It makes use of a few basic ideas from the field of thermo - dynamics, particularly the concept of free energy.
- Changes in free energy (ΔG) provide a measure of the energetic feasibility of a chemical reaction and can, therefore, allow prediction of whether a reaction or process can take place.
- In short, bio - energetics predicts if a process is possible, whereas kinetics measures how fast the reaction occurs

FREE ENERGY

Direction and extent to which a chemical reaction proceeds is determined by the degree to which two factors change during the reaction.

ΔG , represents the change in free energy at any specified concentration of products and reactants.
The standard free energy change, ΔG^o is the energy change when reactants and products are at a concentration of 1 mol/L. [Note: The concentration of protons is assumed to be 10–7 mol/L, that is, pH = 7.] Furthermore, ΔGo can readily be determined from measurement of the equilibrium constant

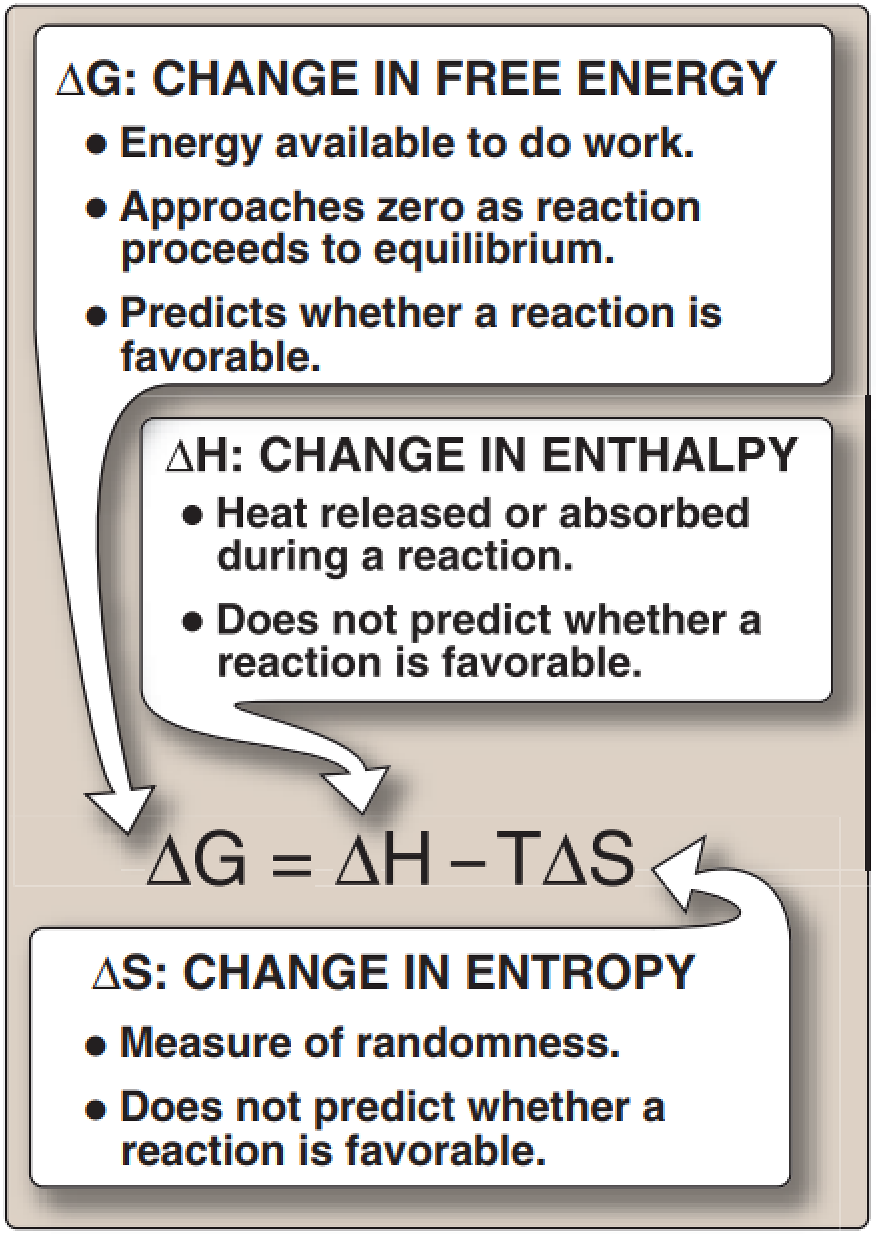
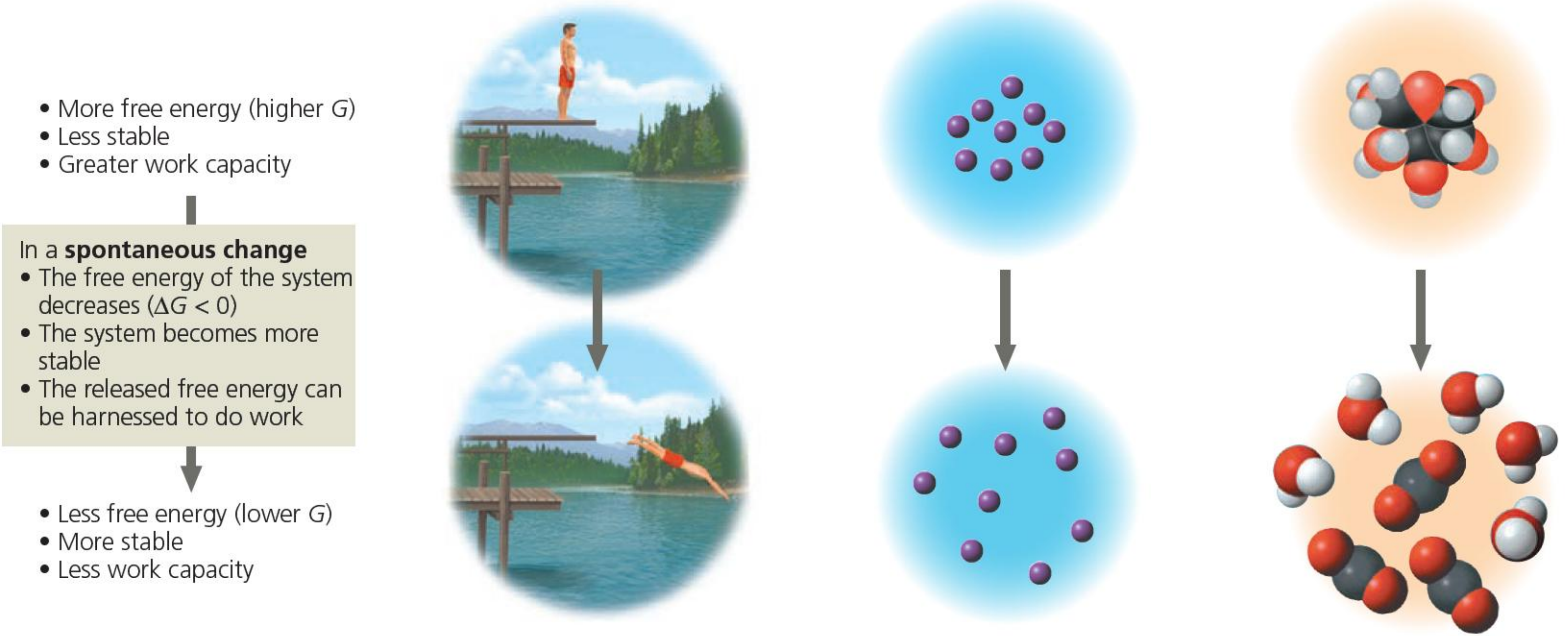


Figure 8.5 The relationship of free energy to stability, work capacity, and spontaneous change. Unstable systems (top) are rich in free energy, *G*. They have a tendency to change spontaneously to a more stable state (bottom), and it is possible to harness this "downhill" change to perform work.



(a) Gravitational motion. Objects move spontaneously from a higher altitude to a lower one.

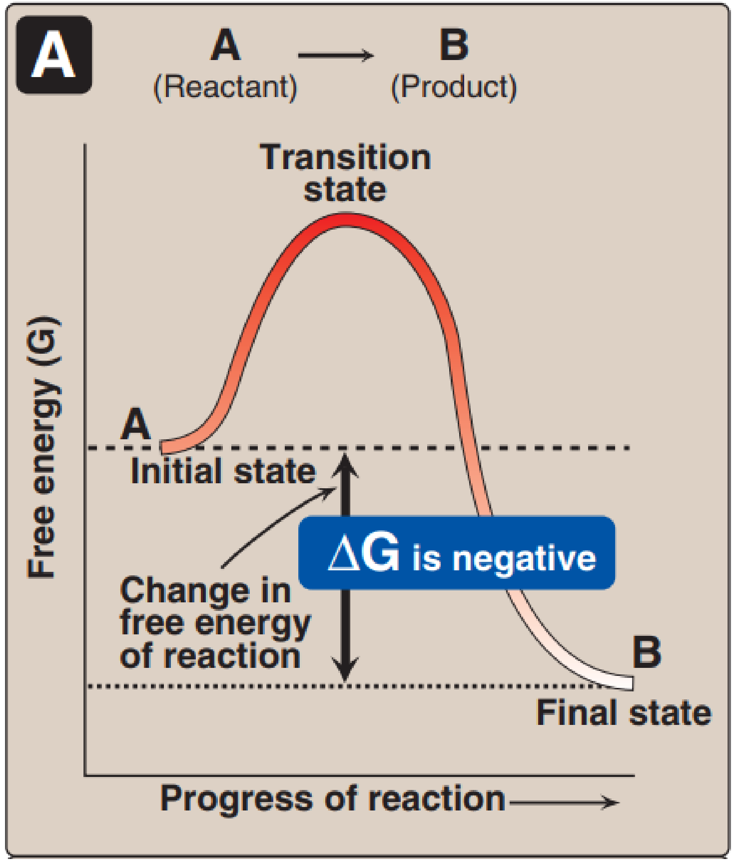
(b) Diffusion. Molecules in a drop of dye diffuse until they are randomly dispersed.

(c) Chemical reaction. In a cell, a glucose molecule is broken down into simpler molecules.

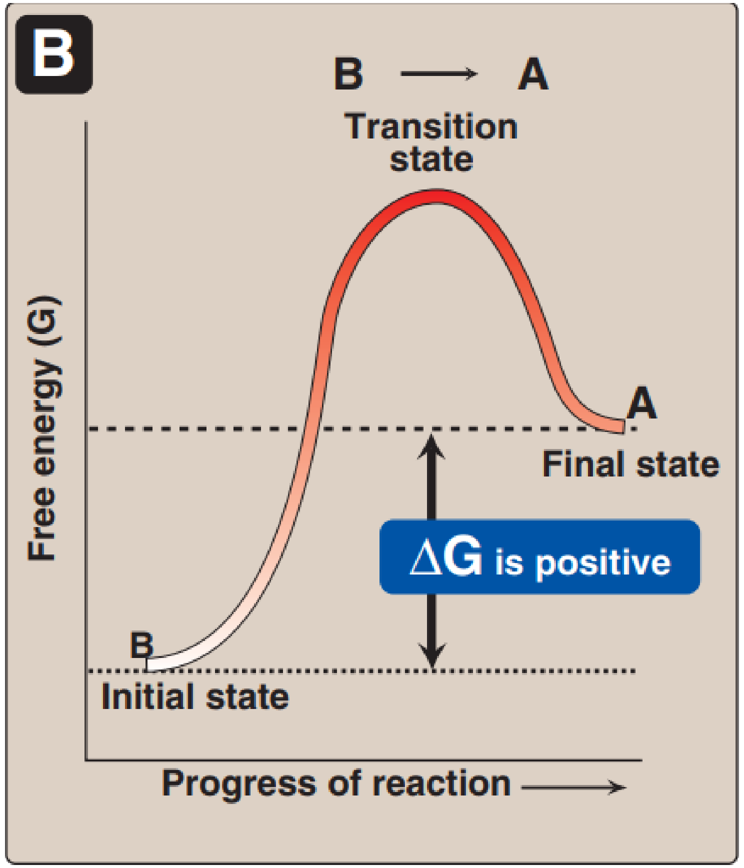
FREE ENERGY CHANGE

If ΔG = 0, the reactants are in equilibrium

exergonic



endergonic



The free energy of the forward reaction ($A \rightarrow B$) is equal in magnitude but opposite in sign to that of the back reaction ($B \rightarrow A$)

$$\boxed{\Delta G = \Delta G^o + RT \ln \frac{[B]}{[A]}}$$

where ΔG^o is the standard free energy change (see below)

R is the gas constant (1.987 cal/mol · degree)

T is the absolute temperature (^oK)

[A] and [B] are the actual concentrations of the reactant and product

ln represents the natural logarithm

A reaction with a positive ΔGo can proceed in the forward direction (have a negative overall ΔG) if the ratio of products to reactants ([B]/[A]) is sufficiently small (that is, the ratio of reactants to products is large).